

Siamese DINO Adapter Learning with Haze-Robust Feature Representation for Nighttime Image Dehazing

Jinhee Park^{1,2} Choongsang Cho¹ Guisik Kim^{1*}

¹Korea Electronics Technology Institute, Seongnam, South Korea

²Chung-Ang University, Seoul, South Korea

iv4084em@keti.re.kr ideafisher@keti.re.kr specialre@naver.com

Abstract

Nighttime image dehazing is a challenging restoration problem due to severe haze, glow, non-uniform illumination, and strong localized light sources. While recent restoration networks provide strong low-level reconstruction capability, they often lack explicit high-level semantic guidance, which is important for preserving scene structure under heavy nighttime degradation. In this work, we investigate how semantic priors from a pretrained vision foundation model can be effectively incorporated into a nighttime dehazing pipeline. To this end, we propose a semantic-guided dual-branch framework that combines a frozen DINOv3 semantic encoder with a NAFNet-based restoration encoder. Our core idea is to explicitly bridge the representational gap between semantically rich foundation features and reconstruction-oriented restoration features, enabling high-level priors to be effectively integrated into a low-level restoration backbone. In addition, we employ lightweight feature adaptation, semantic alignment, and a light-aware synthetic nighttime haze dataset to improve robustness under the limited training setting. Experimental results on the NTIRE 2026 Nighttime Image Dehazing Challenge benchmark show that the proposed method achieves strong and well-balanced performance for nighttime image dehazing.

1. Introduction

Nighttime image dehazing is a challenging low-level vision task that seeks to recover clean scene content from images degraded by haze under complex nighttime illumination [1, 22]. Compared with daytime dehazing [11, 12, 18, 20], nighttime restoration is significantly more difficult due to the coexistence of scattering effects, glow artifacts, severe contrast degradation, and strong localized light sources [6, 10]. These factors not only corrupt low-level image statistics but also obscure structural cues that are impor-

*Corresponding author.

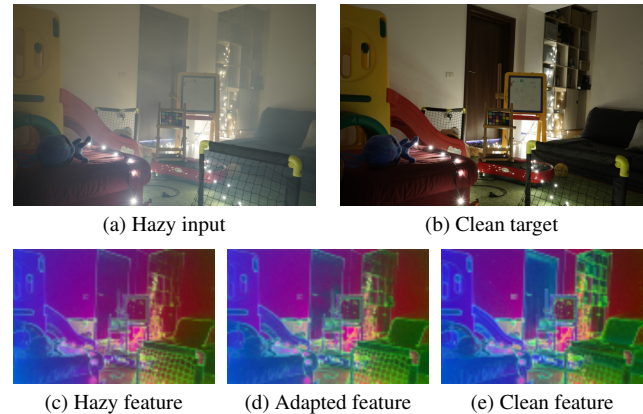


Figure 1. **Motivation for semantic feature adaptation under nighttime haze.** Even under severe haze, the pretrained DINO feature space still preserves coarse scene structure and object-level cues. However, the feature extracted from the hazy image is not fully aligned with the clean-image feature. After passing through the proposed adapter, the feature becomes more consistent with the clean semantic representation, motivating our feature-level adaptation and Siamese semantic alignment design. For visualization, intermediate token features are projected to RGB using the top three PCA components.

tant for faithful scene recovery, making nighttime dehazing challenging.

Recent image restoration networks have achieved strong performance by improving local reconstruction capability through convolutional or transformer-based architectures [4, 13, 21]. However, most existing pipelines remain primarily low-level in nature: they focus on pixel-level restoration and local feature processing, while only implicitly modeling high-level semantics. Such designs are effective when degradation can be addressed largely through local reconstruction, but may be insufficient for nighttime dehazing, where severe haze distorts not only textures and contrast but also semantically meaningful structure.

Meanwhile, large-scale pretrained vision foundation models provide rich and transferable semantic representa-

tions across visual tasks. Recent restoration works have also suggested that such pretrained semantic priors can be useful for low-level vision tasks [14, 24]. In particular, DINOv3 [16] produces strong dense visual features that capture scene structure across multiple representation depths, suggesting that high-level semantic priors from a frozen foundation model could help preserve scene structure under severe nighttime degradation. To better examine this possibility, we visualize intermediate DINO features for clean and hazy nighttime images. As illustrated in Fig. 1, even under severe haze, the pretrained representation still preserves coarse object layout and scene structure, indicating that useful information behind the haze remains encoded in the frozen DINOv3 feature space.

This observation suggests that strong pretrained features can provide meaningful semantic priors even for degraded nighttime inputs. However, directly injecting such features into a low-level restoration network is non-trivial. The semantic features from a pretrained transformer are globally structured and representation-oriented, whereas restoration features are reconstruction-oriented and strongly tied to spatial detail. As a result, naive fusion is unlikely to fully exploit the potential of pretrained semantic priors.

In this paper, we address this issue by asking a simple but important question: *how can high-level semantic priors be made compatible with a low-level restoration pipeline for nighttime dehazing?* To answer this, we propose a dual-branch framework that combines a frozen DINOv3 semantic encoder with a NAFNet [4]-based restoration encoder. Our core idea is not merely to append DINOv3 features to a restoration backbone, but to explicitly bridge the representational gap between semantic and restoration features.

Specifically, we introduce lightweight residual bottleneck adapters on selected intermediate DINOv3 token features, allowing haze-corrupted nighttime inputs to be corrected at the feature level without altering the frozen pretrained backbone. We then design a semantic-to-restoration bridge that converts the adapted DINOv3 features into a restoration-compatible representation for fusion with the restoration backbone. To further regularize the semantic branch, we employ a Siamese alignment loss between hazy adapted features and clean-image features extracted from the same frozen DINOv3 backbone. This encourages degraded semantic features to remain aligned with the clean semantic space during restoration.

Based on this design, our framework converts high-level semantic priors into a restoration-compatible bottleneck representation that can be effectively processed by standard low-level restoration components. This semantically enriched bottleneck feature is further refined by Restormer [21]-based blocks and progressively decoded by a NAFNet decoder, allowing the network to benefit from both semantic guidance and restoration-oriented recon-

struction. In this way, our method preserves the strengths of established low-level vision pipelines while enhancing them with semantically informed representations. To further improve generalization under the limited training set, we additionally construct a light-aware synthetic nighttime haze dataset that explicitly models localized light sources and depth-dependent scattering.

Our main contributions are summarized as follows:

- We propose a dual-branch nighttime dehazing framework that incorporates frozen DINOv3 semantic priors into a low-level restoration backbone.
- We introduce lightweight residual bottleneck adapters and a Siamese semantic alignment loss to preserve the pretrained semantic structure while adapting frozen DINOv3 features to haze-corrupted nighttime inputs.
- We design a semantic-to-restoration bridge that converts semantically rich DINOv3 features into a restoration-compatible bottleneck representation.

2. Related Work

2.1. Nighttime Dehazing

Dehazing has been extensively studied in the daytime setting. Early deep learning methods, such as AOD-Net [11], formulated dehazing as an end-to-end image restoration problem, while later approaches improved reconstruction quality using multi-scale designs, attention mechanisms, contrastive learning, and frequency-aware modeling [12, 18, 20]. These methods have significantly advanced daytime dehazing, where degradations are often modeled primarily by atmospheric scattering and contrast attenuation.

Nighttime dehazing is considerably more challenging. In addition to haze scattering, nighttime scenes often contain strong localized light sources, glow artifacts, non-uniform illumination, and severe contrast degradation, making restoration much harder than in the daytime setting [6, 10, 22]. Recent nighttime dehazing methods have therefore introduced more specialized architectures and training strategies to handle these effects [6, 13]. Nevertheless, most methods remain largely low-level in nature, focusing on pixel reconstruction and local feature processing. While effective for recovering visibility, they often lack explicit mechanisms for leveraging semantic priors that may help preserve structure under severe nighttime degradation.

2.2. Semantic-guided Image Restoration

To improve restoration beyond purely low-level reconstruction, recent works have explored incorporating high-level representations into image restoration pipelines. One line of research leverages pretrained vision foundation models as auxiliary guidance. For example, DA-CLIP [14] introduces a degradation-aware controller to adapt CLIP features for multi-task image restoration, while Perceive-IR [24] em-

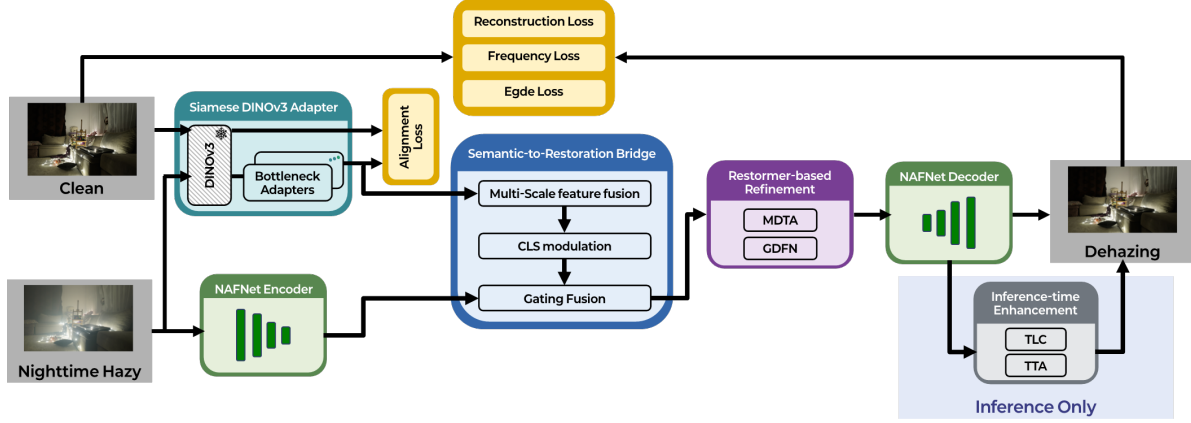


Figure 2. **Overview of the proposed nighttime dehazing framework.** The hazy input is processed by a frozen DINOv3[16] semantic encoder and a NAFNet[4] restoration encoder in parallel. Multi-scale semantic features are extracted from selected transformer layers and refined using lightweight residual bottleneck adapters. These features are transformed into a restoration-compatible representation through the proposed semantic-to-restoration bridge, which consists of multi-layer feature fusion, CLS-based global prior modulation, and adaptive gating fusion with the NAFNet encoder. The fused latent representation is further refined by Restormer [21]-based refinement and decoded by a NAFNet decoder. During inference, additional robustness is achieved via Test-time Local Converter [5] and test-time augmentation.

employs semantic guidance from pretrained large-scale vision models to improve all-in-one restoration. These works suggest that semantic priors can benefit low-level restoration.

However, they are mainly developed for general or multi-task restoration settings, and typically use semantic features as auxiliary guidance. In contrast, our method explicitly addresses the representational gap between semantically structured foundation features and reconstruction-oriented restoration features for nighttime dehazing. To this end, we design a semantic-to-restoration bridge and further employ lightweight feature adaptation and Siamese semantic alignment to preserve pretrained semantic structure under haze-corrupted nighttime inputs.

3. Proposed Method

Our goal is not merely to use DINOv3[16] features for nighttime dehazing, but to make high-level semantic priors compatible with a low-level restoration pipeline. As shown in Fig. 2, we propose a dual-branch architecture composed of a frozen DINOv3 semantic encoder and a NAFNet[4] restoration encoder. The key component of our framework is a semantic-to-restoration bridge that transforms semantically rich but restoration-misaligned DINOv3 features into a representation that can be effectively fused with reconstruction-oriented bottleneck features. In addition, we impose a Siamese semantic alignment loss so that haze-corrupted semantic features remain close to clean-image representations in the frozen DINOv3 feature space.

3.1. Frozen DINOv3 Semantic Encoder

We use a frozen DINOv3 backbone [16] to provide high-level semantic priors. Instead of fine-tuning the backbone,

we extract intermediate token features from four tap layers, namely layers 5, 11, 17, and 23, and apply lightweight residual bottleneck adapters only to these tapped features. This design preserves the pretrained semantic representation while enabling feature-level correction for haze-corrupted nighttime inputs.

Let $F_l \in \mathbb{R}^{N \times D}$ denote the patch-token feature from tap layer l . The adapted feature is defined as

$$\tilde{F}_l = A_l(F_l) = F_l + \Delta_l(F_l), \quad (1)$$

where $A_l(\cdot)$ is a lightweight residual bottleneck adapter. Importantly, the adapters are applied only after the forward pass and do not modify the internal transformer propagation. Thus, the DINOv3 backbone remains fully frozen, and only a lightweight feature-level correction is learned.

3.2. NAFNet Restoration Encoder

In parallel with the semantic branch, we employ a NAFNet [4]-based encoder to extract restoration-oriented features from the hazy input image. Unlike the DINOv3 features, which are semantically structured and globally informative, the NAFNet encoder features are optimized for local reconstruction and spatial detail recovery. These two branches therefore provide complementary information: the DINOv3 branch contributes semantic guidance, while the NAFNet branch provides low-level restoration features.

3.3. Semantic-to-Restoration Bridge

The central challenge of our framework is to bridge the representational gap between high-level semantic features and low-level restoration features. To address this issue, we design a semantic-to-restoration bridge composed of

three steps: multi-layer semantic feature fusion, CLS-based global prior modulation, and adaptive gating fusion.

Multi-layer semantic feature fusion. The adapted token features from the four tap layers capture complementary semantic information at different transformer depths. To integrate them, we first reshape each token sequence into a 2D feature map and project each map into a shared channel space using a layer-specific 1×1 convolution. The projected features are then concatenated along the channel dimension and refined by stacked 3×3 convolutions:

$$F_{\text{dino}} = \phi([\psi_1(\tilde{F}_{l_1}), \psi_2(\tilde{F}_{l_2}), \psi_3(\tilde{F}_{l_3}), \psi_4(\tilde{F}_{l_4})]), \quad (2)$$

where $\psi_i(\cdot)$ aligns the feature of the i -th layer to a shared embedding space, and $\phi(\cdot)$ denotes convolutional fusion over the concatenated multi-level features. This produces a unified multi-scale semantic feature map.

CLS-based global prior modulation. In addition to dense token features, DINOv3 provides a class token that summarizes the global image content. We transform this token into a channel-wise modulation vector and use it to modulate the fused semantic feature:

$$s = \sigma(\text{MLP}(c)), \quad \tilde{F}_{\text{dino}} = F_{\text{dino}} \odot s, \quad (3)$$

where c is the class token and $\odot$ denotes channel-wise multiplication with broadcasting. This operation injects global scene context into the dense semantic representation.

Adaptive gating fusion. After semantic fusion and global modulation, the DINOv3 feature is fused with the NAFNet bottleneck feature through an adaptive gate. Let F_{naf} denote the NAFNet bottleneck feature and $\tilde{F}_{\text{dino}}$ denote the modulated semantic prior. We compute

$$G = \sigma(W_n * F_{\text{naf}} + W_d * \tilde{F}_{\text{dino}}), \quad (4)$$

and obtain the fused bottleneck feature as

$$F_{\text{fused}} = G \odot F_{\text{naf}} + (1 - G) \odot \tilde{F}_{\text{dino}}. \quad (5)$$

This gate adaptively balances reconstruction-oriented evidence and semantic guidance, making frozen DINOv3 features compatible with the restoration backbone.

3.4. Bottleneck Refinement and Decoder

The fused representation still contains heterogeneous information from the semantic and restoration branches. We therefore refine the fused bottleneck feature using Restormer blocks [21], which consist of multi-Dconv head transposed attention (MDTA) and gated-Dconv feed-forward networks (GDFN), to enhance long-range context

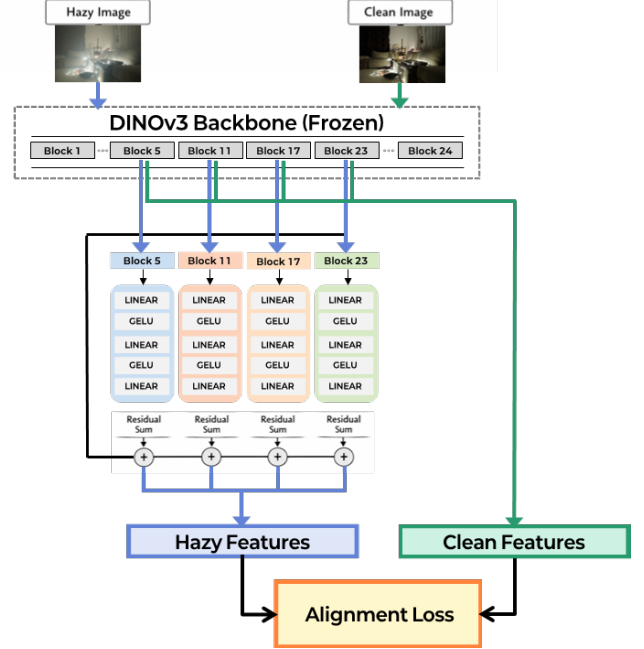


Figure 3. **Frozen DINOv3 semantic encoder with Siamese alignment.** Intermediate token features are extracted from selected DINOv3 layers. Adapters are applied only to the hazy branch, and the adapted hazy features are aligned with clean-image features using a Siamese feature loss.

interaction and adaptive feature transformation before decoding. The refined bottleneck feature is then progressively upsampled by a NAFNet decoder. The decoder outputs a residual image, which is added to the input to form the final restored image.

3.5. Training Objective

Training is supervised at both the image and feature levels. The final restored image is optimized using reconstruction, frequency-domain, and edge-preservation losses, while the adapted DINOv3 features are regularized by a Siamese semantic alignment loss. The total training loss is defined as

$$\mathcal{L} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{fft}} \mathcal{L}_{\text{fft}} + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \lambda_{\text{feat}} \mathcal{L}_{\text{feat}}, \quad (6)$$

where $\mathcal{L}_{\text{rec}}$ is an ℓ_1 reconstruction loss, $\mathcal{L}_{\text{fft}}$ is a frequency-domain consistency loss, and $\mathcal{L}_{\text{edge}}$ is a Sobel-based edge preservation loss.

Our main semantic regularization term is the Siamese feature alignment loss, which is illustrated in Fig. 3. During training, the clean target image is passed through the same frozen DINOv3 backbone without adapters, producing clean semantic features F_l^{gt} . The adapted hazy features $\tilde{F}_l^h$ are then aligned with the clean features as

$$\mathcal{L}_{\text{feat}} = \frac{1}{|\mathcal{S}|} \sum_{l \in \mathcal{S}} \|\tilde{F}_l^h - F_l^{\text{gt}}\|_1, \quad (7)$$

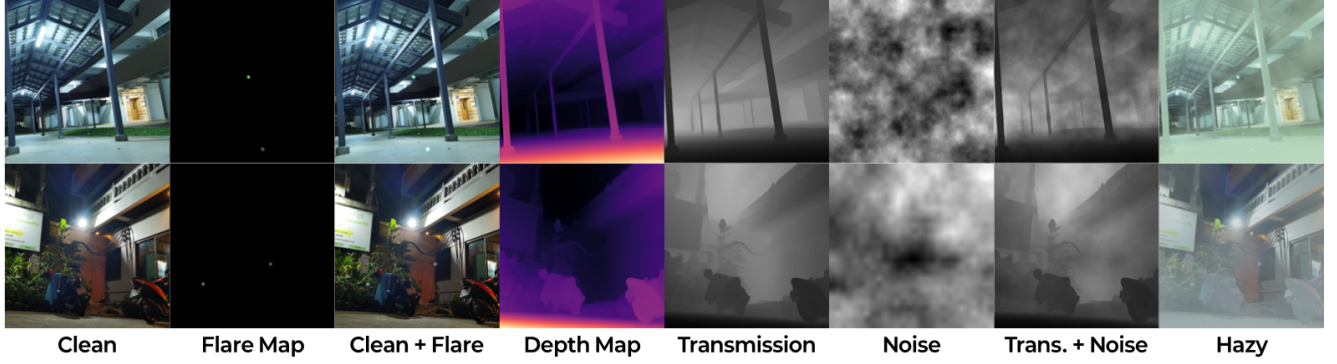


Figure 4. **Step-by-step visualization of the proposed nighttime haze synthesis pipeline.** (a) clean nighttime image, (b) sampled flare map, (c) flare-composited clean image, (d) estimated depth map, (e) transmission map derived from depth, (f) spatial noise map, (g) noisy transmission used for non-uniform haze synthesis, and (h) final synthesized hazy image.

where $\mathcal{S} = \{5, 11, 17, 23\}$ is the set of tap layers. This loss encourages haze-corrupted semantic features to remain consistent with the clean semantic space while preserving the pretrained DINOv3 representation.

3.6. Inference-time Enhancement

Our model is trained on random crops of size 512×512 , whereas the challenge test images have much higher resolution (e.g., 1800×1200). This train-test resolution gap can lead to a distribution mismatch, especially when global aggregation modules are involved. To alleviate this issue, we apply Test-time Local Converter (TLC) [5] during inference, which makes the inference behavior more consistent with the local statistics observed during patch-based training. In addition, we apply test-time augmentation (TTA) to further improve prediction robustness. These techniques are used only during inference.

3.7. Extra Dataset and Nighttime Haze Synthesis

The challenge training set is relatively limited, which makes it difficult to learn a robust nighttime dehazing model solely from the provided data. To improve generalization, we construct an additional synthetic training set tailored to the characteristics of nighttime haze.

A key observation is that nighttime hazy scenes often contain strong localized light sources, whose glow and scattering effects significantly influence the appearance of haze. To reflect this property, we build a light-aware nighttime haze synthesis pipeline based on Flare7K++ [7]. As illustrated in Fig. 4, we first sample a nighttime image and a light-source image, and composite the light source onto the nighttime image to obtain a flare-aware clean image. We then estimate its depth map using Depth Anything V2[19] and derive a transmission map from the estimated depth using the standard atmospheric scattering model:

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (8)$$

where $J(x)$ denotes the flare-aware clean image, $t(x)$ is the transmission map, and A is the atmospheric light. To better reflect the non-uniform nature of nighttime haze, the transmission is further perturbed with spatial noise. Finally, the hazy image is synthesized using the atmospheric scattering model with additional grain noise. Through this process, we generate synthetic nighttime hazy images that better match the challenge setting, particularly in terms of localized light sources and glow-aware haze appearance.

4. Experiments

4.1. Experimental Setup

We implement our method in Python using PyTorch and train it on a single NVIDIA RTX PRO 6000 GPU. Training is conducted in two stages with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and cosine annealing learning-rate schedules. In the first stage, the model is trained on random 512×512 patches with batch size 4 using both the challenge data and the synthesized nighttime haze dataset, with the learning rate annealed from 1×10^{-4} to 1×10^{-6} . In the second stage, the model is fine-tuned on challenge data only using 768×768 random patches with batch size 1, with the learning rate annealed from 5×10^{-6} to 1×10^{-6} . The loss weights are set to $\lambda_{rec} = 1.0$, $\lambda_{fft} = 0.05$, $\lambda_{edge} = 0.1$, and $\lambda_{feat} = 0.04$. The total training time is approximately 20 hours. At inference time, we use Test-time Local Converter (TLC) [5] and 8-way Test-time augmentation (TTA), and the average runtime is 5.25 seconds per image.

4.2. Synthetic Data Generation Details

For synthetic nighttime haze generation, we randomly sample a nighttime image and a light-source image from Flare7K++ [7]. The light source is resized to 35%–65% of its original size and composited 4–7 times in the upper 30% region of the image with random alpha values between 0.5 and 0.9. Depth is estimated using Depth Any-



Figure 5. **Examples from the synthesized nighttime haze dataset.** The top row shows clean nighttime images used as the synthesis base, and the bottom row shows the corresponding synthetic hazy images generated by our light-aware nighttime haze synthesis pipeline.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MUSIQ $\uparrow$	NIQE $\downarrow$	FID $\downarrow$	Final Rank
Team A	27.880 (1)	<u>0.905</u> (2)	0.101 (1)	60.395 (12)	3.771 (5)	<u>72.250</u> (2)	1
Team B	<u>27.440</u> (2)	0.906 (1)	0.110 (4)	63.264 (7)	3.990 (12)	64.731 (1)	2
Ours	27.300 (3)	0.903 (3)	0.109 (3)	64.051 (3)	<u>3.680</u> (3)	74.466 (3)	3
Team C	26.843 (4)	0.897 (4)	<u>0.106</u> (2)	<u>64.542</u> (2)	3.644 (1)	90.653 (7)	4
Team D	26.626 (5)	0.893 (5)	0.117 (6)	65.253 (1)	3.951 (7)	86.058 (4)	5

Table 1. **Anonymized top-5 leaderboard results on the NTIRE 2026 Nighttime Dehazing Challenge benchmark.** Numbers in parentheses denote the rank of each method under the corresponding metric among all 21 participating teams. For blind review, all competing team identities are anonymized, and the row corresponding to our method is highlighted in gray. The best result for each metric is shown in **bold**, and the second-best result is underlined.

thing V2-Small [19], and the transmission is computed as $t(x) = \exp(-\beta d(x))$ with $\beta \in [2.5, 5.0]$, followed by additional base fog and 4-octave Perlin noise for non-uniform haze. Atmospheric light intensity is sampled from 0.65 to 0.95 and adjusted according to the dominant color of the inserted light source. Finally, grain noise with standard deviation 0.005–0.015 is added. Examples of the synthesized data are shown in Fig. 5.

4.3. Nighttime Image Dehazing Challenge dataset

We evaluate our method on the NTIRE 2026 Nighttime Image Dehazing Challenge dataset [2]. The dataset consists of indoor nighttime hazy scenes containing dense localized light sources, such as small decorative bulbs arranged in series, which induce strong glow and scattering effects. These characteristics make the benchmark particularly challenging, as the model must remove haze while preserving fine structures and handling several small bright light sources.

The dataset provides 25 paired training images, 5 paired validation images, and 5 test images, all at a resolution of 1800×1200 . The validation and test sets share similar indoor nighttime haze characteristics with the training set, in-

cluding dense small light sources and strong glow effects.

4.4. Main Challenge Results

Table 1 reports the anonymized top-5 leaderboard results on the NTIRE 2026 Nighttime Dehazing Challenge benchmark [3]. The benchmark evaluates restoration quality using both distortion-based and perceptual metrics. Specifically, PSNR and SSIM [17] measure pixel-level reconstruction fidelity and structural similarity, respectively, while LPIPS [23] evaluates perceptual similarity. NIQE [15] and MUSIQ [9] assess perceptual naturalness and image quality without reference, and FID [8] measures distributional similarity in feature space.

Our method achieved a final rank of 3rd overall under the official challenge evaluation protocol. Notably, while some competing methods perform better on individual metrics, our approach maintains consistently competitive ranks across all reported criteria among all 21 participating teams. In particular, our method does not exhibit a severe drop on any single metric, suggesting that it achieves a well-balanced trade-off between reconstruction fidelity, perceptual quality, and visual naturalness.



Figure 6. **Qualitative cumulative ablation results on the validation set.** Each row shows one validation sample, and the columns correspond to *Input*, *NAFNet*, *+ DINOv3*, *+ Siamese alignment*, *+ Restormer-based refinement*, and *+ fine-tuning with 768×768 patches*, respectively. As more components are progressively added, the restored results exhibit clearer haze removal, improved structure preservation, and better handling of localized light sources and glow.

We believe this balanced performance is closely related to the proposed design. By explicitly integrating DINOv3 semantic priors into a low-level restoration pipeline, the proposed framework improves restoration quality without over-optimizing for only one aspect of image quality. These results support the effectiveness of our semantic-guided nighttime dehazing framework in the challenge setting.

4.5. Ablation Study

We conduct ablation studies on an internal hold-out split constructed from the training set. Specifically, images #1–20 are used for training, while images #21–25 are excluded from training and used only for PSNR evaluation. Table 2 presents a cumulative ablation study under this setting.

In Stage 1, starting from the NAFNet baseline, adding the synthesized nighttime haze dataset yields a modest gain. Introducing the DINOv3 semantic branch brings a substantial improvement, confirming that high-level semantic priors are beneficial for nighttime dehazing. Adding Siamese alignment further improves PSNR, indicating that aligning degraded semantic features with clean-image representations helps stabilize semantic guidance. Applying the proposed Restormer-based refinement yields another gain, sug-

Stage	Configuration	PSNR $\uparrow$
1	NAFNet (w/o extra data)	24.93
1	NAFNet (+ extra data)	25.04
1	+ DINOv3 encoder	27.38
1	+ Siamese alignment	27.93
1	+ Restormer-based Refinement	28.26
2	+ Fine-tuning with 768×768 patch	28.74

Table 2. **Cumulative ablation study on an internal hold-out split.** We progressively add the proposed components starting from the NAFNet [4] baseline. Stage 1 uses 512×512 random patches, while Stage 2 further fine-tunes the full model using 768×768 random patches.

gesting that semantically enriched bottleneck features benefit from additional global refinement.

In Stage 2, we further fine-tune the full model using larger 768×768 random patches, which provides the best result. This suggests that higher-resolution adaptation is important for the target nighttime dehazing setting.

Fig. 6 shows qualitative cumulative ablation results on five validation samples. Starting from the NAFNet baseline,

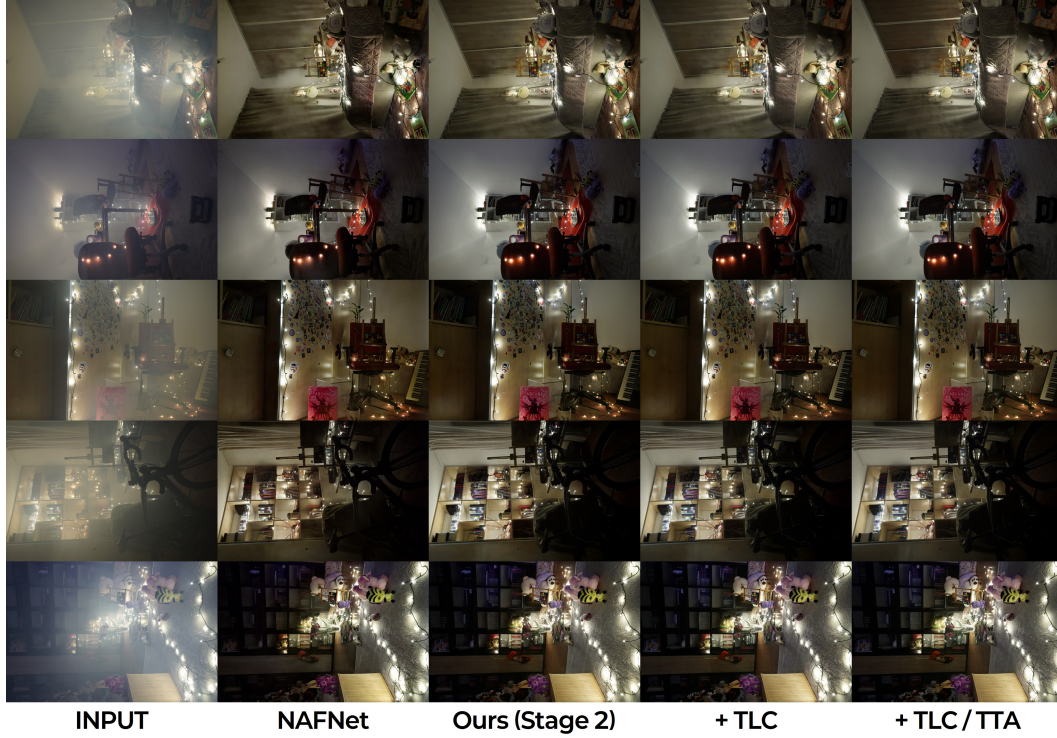


Figure 7. **Qualitative effect of inference-time enhancement.** Each row shows one challenge test sample, and the columns correspond to *Input*, *NAFNet*, *Ours(Stage-2)*, *+ TLC*, and *+ TLC + TTA*, respectively. Compared with the NAFNet baseline, the proposed Stage-2 model produces clearer dehazing results and better preserves scene structure. Applying TLC improves consistency under full-resolution inference, while the additional use of TTA produces the most visually stable and refined results.

progressively adding the proposed components consistently improves visual quality. In particular, introducing the DINOv3 branch leads to better scene consistency and more faithful recovery of semantically meaningful structures. Adding Siamese alignment further stabilizes the restoration results, especially in regions affected by severe haze and glow. The proposed Restormer-based refinement produces cleaner and more coherent outputs, while the final Stage-2 fine-tuning with larger 768×768 patches further improves overall sharpness and visual realism. Overall, the qualitative results are consistent with the quantitative gains in Table 2, showing that the proposed components progressively enhance nighttime dehazing quality.

4.5.1. Inference-time Enhancement

Fig. 7 illustrates the effect of inference-time enhancement on challenge test images. Compared with the NAFNet [4] baseline, the proposed Stage-2 model already produces clearer dehazing results and better restores challenging nighttime structures. Applying Test-time Local Converter (TLC) [5] further improves restoration consistency under full-resolution inference by reducing the discrepancy between patch-based training and large-resolution testing. Finally, adding test-time augmentation (TTA) on top of TLC

yields the most visually stable results, with cleaner haze removal and more coherent overall appearance.

5. Conclusion

In this paper, we presented a semantic-guided framework for nighttime image dehazing that incorporates high-level priors from a frozen DINOv3 backbone into a low-level restoration pipeline. Our core idea was to explicitly address the representational gap between semantically rich foundation features and reconstruction-oriented restoration features. To this end, we developed a semantic-guided integration framework that makes foundation features compatible with a reconstruction-oriented nighttime dehazing pipeline.

Built upon these components, the proposed framework preserves the strengths of established low-level restoration models while benefiting from semantically informed guidance. We further improved robustness under the challenge setting through a light-aware synthetic nighttime haze dataset and inference-time enhancement strategies. Experimental results on the NTIRE 2026 Nighttime Image Dehazing Challenge benchmark demonstrated that the proposed method achieves strong and well-balanced performance across multiple evaluation metrics.

Acknowledgments

This work was partly supported by Institute of Information communications Technology Planning Evaluation (IITP) grant funded by the Korea government (MSIT) (No.RS-2022-II220926, RS-2022-II220984, RS-2024-00456709)

References

- [1] Cosmin Ancuti, Codruta O Ancuti, Christophe De Vleeschouwer, and Alan C Bovik. Day and night-time dehazing by local airlight estimation. *IEEE Transactions on Image Processing*, 29:6264–6275, 2020. 1
- [2] Radu Ancuti, Codruta Ancuti, Radu Timofte, and Cosmin Ancuti. NT-HAZE: A Benchmark Dataset for Realistic Night-time Image Dehazing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2026. 6
- [3] Radu Ancuti, Alexandru Brateanu, Florin Vasluianu, Raul Balmez, Ciprian Orhei, Codruta Ancuti, Radu Timofte, Cosmin Ancuti, et al. NTIRE 2026 Nighttime Image Dehazing Challenge Report. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2026. 6
- [4] Liangyu Chen, Xiaojie Chu, Xiangyu Zhang, and Jian Sun. Simple baselines for image restoration. In *European conference on computer vision*, pages 17–33. Springer, 2022. 1, 2, 3, 7, 8
- [5] Xiaojie Chu, Liangyu Chen, Chengpeng Chen, and Xin Lu. Improving image restoration by revisiting global information aggregation. In *European Conference on Computer Vision*, pages 53–71. Springer, 2022. 3, 5, 8
- [6] Xiaofeng Cong, Jie Gui, Jing Zhang, Junming Hou, and Hao Shen. A semi-supervised nighttime dehazing baseline with spatial-frequency aware and realistic brightness constraint. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2631–2640, 2024. 1, 2
- [7] Yuekun Dai, Chongyi Li, Shangchen Zhou, Ruicheng Feng, Yihang Luo, and Chen Change Loy. Flare7k++: Mixing synthetic and real datasets for nighttime flare removal and beyond. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(11):7041–7055, 2024. 5
- [8] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017. 6
- [9] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5148–5157, 2021. 6
- [10] Shiba Kuanar, Dwarikanath Mahapatra, Monalisa Bilas, and KR Rao. Multi-path dilated convolution network for haze and glow removal in nighttime images. *The Visual Computer*, 38(3):1121–1134, 2022. 1, 2
- [11] Boyi Li, Xiulian Peng, Zhangyang Wang, Jizheng Xu, and Dan Feng. Aod-net: All-in-one dehazing network. In *Proceedings of the IEEE international conference on computer vision*, pages 4770–4778, 2017. 1, 2
- [12] Xiaohong Liu, Yongrui Ma, Zhihao Shi, and Jun Chen. Grid-dehazenet: Attention-based multi-scale network for image dehazing. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7314–7323, 2019. 1, 2
- [13] Yun Liu, Zhongsheng Yan, Sixiang Chen, Tian Ye, Wenqi Ren, and Erkang Chen. Nighthazeformer: Single nighttime haze removal using prior query transformer. In *Proceedings of the 31st ACM international conference on multimedia*, pages 4119–4128, 2023. 1, 2
- [14] Ziwei Luo, Fredrik K Gustafsson, Zheng Zhao, Jens Sjölund, and Thomas B Schön. Controlling vision-language models for multi-task image restoration. *arXiv preprint arXiv:2310.01018*, 2023. 2
- [15] Anish Mittal, Rajiv Soundararajan, and Alan C Bovik. Making a “completely blind” image quality analyzer. *IEEE Signal processing letters*, 20(3):209–212, 2012. 6
- [16] Oriane Siméoni, Huy V Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025. 2, 3
- [17] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004. 6
- [18] Haiyan Wu, Yanyun Qu, Shaohui Lin, Jian Zhou, Ruizhi Qiao, Zhizhong Zhang, Yuan Xie, and Lizhuang Ma. Contrastive learning for compact single image dehazing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10551–10560, 2021. 1, 2
- [19] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything v2. *Advances in Neural Information Processing Systems*, 37:21875–21911, 2024. 5, 6
- [20] Hu Yu, Naishan Zheng, Man Zhou, Jie Huang, Zeyu Xiao, and Feng Zhao. Frequency and spatial dual guidance for image dehazing. In *European conference on computer vision*, pages 181–198. Springer, 2022. 1, 2
- [21] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, and Ming-Hsuan Yang. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5728–5739, 2022. 1, 2, 3, 4
- [22] Jing Zhang, Yang Cao, Zheng-Jun Zha, and Dacheng Tao. Nighttime dehazing with a synthetic benchmark. In *Proceedings of the 28th ACM international conference on multimedia*, pages 2355–2363, 2020. 1, 2
- [23] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. 6

- [24] Xu Zhang, Jiaqi Ma, Guoli Wang, Qian Zhang, Huan Zhang, and Lefei Zhang. Perceive-ir: Learning to perceive degradation better for all-in-one image restoration. *IEEE Transactions on Image Processing*, 2025. [2](#)